

What are the problems of old methods in Generative Modelling (GM) ?

1. Hard assumptions about structure of the data.
2. Too much relaxation of the models.
3. Relying on high amount of computation in inference using monte carlo methods.

VAE: Assumptions about structure of data is weak, its training and inference is fast. (VAEs do make an approximation but the approximation is arguably small.)

Latent Variable Models:

Recall in generative models we aim to capture dependencies between pixels of an image. Now lets take MNIST as an example. Assume that we have generated the first half of an image now we want to generate the second half, for instance if we generate the first half of pixels for digit 5, those pixels are not compatible to any other pixel inside the image. **Thus if we know what we are generating the problem is much easier.**

Let's make things a little formal...

For every point x in sample there should be at least one or many settings of latent variable that causes the model to generate something very similar to x .

Formally we have a latent variable z where we can sample from a high dimensional space Z easily, Then we have a family of DETERMINISTIC function $f(z; \theta)$ where we can map our latents to X space.

our goal is to optimize θ such that once we sample z from $P(z)$ then mapping $f(z; \theta)$ have high probability!

$$\text{sample } z \xrightarrow{\text{map } \theta} f(z; \theta) \sim x$$

There is a z or " z "s that this f has high prob. in x space. So we update θ such that this high prob occurs.

$$P(x) = \int f(z; \theta) P(z) dz$$

or we can say $f(z; \theta) = P(x|z; \theta)$

$$P(x) = \int P(x|z; \theta) P(z) dz$$

in VAEs we assume

$$P(x|z; \theta) \sim \mathcal{N}(x | f(z, \theta), \sigma^2 I)$$

Q: f is deterministic
 z is random $\rightarrow f(z; \theta)$ is deterministic

$\downarrow$
 $P(x|z) \rightarrow$ we also make
a ball around it!
why $P(x|z; \theta)$ is not
determined given z ? $\leftarrow$

if $P(x|z; \theta)$ become dirac delta function then
we would map every point z to a point x
without leaving any thing

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It seems like that the z are sampled randomly at first then mapped to some other form that can help us to generate our digit. Basically the assumption here is that knowing what " z " to use is very hard and cannot be basically random!

So we sample that, map it to a posterior z then sample X from posterior Z through some mappings.

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First, how do we choose the latent variables z such that we capture latent information? Returning to our digits example, the 'latent' decisions that the model needs to make before it begins painting the digit are actually rather complicated. It needs to choose not just the digit, but the angle that the digit is drawn, the stroke width, and also abstract stylistic properties. Worse, these properties may be correlated: a more angled digit may result if one writes faster, which also might tend to result in a thinner stroke. Ideally, we want to avoid deciding by hand what information each dimension of z encodes (although we may want to specify it by hand for some dimensions [4]).

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We also want to avoid explicitly mentioning the dependencies between dimensions of z !

How VAEs Solve this????

They assert that z are sampled from a simple distribution like isotropic Gaussian $z \sim N(0, I)$

* These z 's will be mapped to some latent values using some neural network.

in middle layers!

Now VAEs solved all of the problems that we had.

Basically we wanted to have $P(X|Z, \theta)$:

$$P(X|Z, \theta) = \mathcal{N}(X | f(z, \theta), \sigma^2 I)$$

* where z sampled from isotropic gaussian

we need to have an easy-to-compute $P(X)$

Now what we have:

$$P(X) = \sum_{z_i} P(X|z_i)$$

basically we sample some z_i 's

then compute $P(X|z_i)$.

Basically if we naively sample z , $P(X|z)$ need enormous number of samples to help us to generate good samples. instead VAE sample in a smart way to generate such samples.

We hope space of z is that are lucky under q we have much smaller space. $\Rightarrow$ summation would be easier then.

We can compute $E_{z \sim Q} \{ P(X|z) \}$ much easier!

Goal:

relate $E_{z \sim Q} \{ P(X|z) \}$ and $P(X)$

we start with simple things that we know

We know if we observe X , the $z - P(z|X)$ - should be similar to what Q should have produced - $Q(z)$!

So basically

$$D \{ Q(z) \parallel P(z|X) \}$$

why Q is like?

we want to take expert batch over $Q(z)$

$$= \mathbb{E}_{z \sim Q} \left\{ \log \frac{Q}{P} \right\} = \mathbb{E}_z \left\{ \log Q(z) - \log P(z|x) \right\}$$

$$= \mathbb{E}_z \left\{ \underbrace{\log Q(z)}_{\text{KL}(\ast)} - \log P(x|z) - \underbrace{\log P(z)}_{\text{KL}(\ast)} \right\} + \log P(x)$$

$$= -\mathbb{E}_z \left\{ \log P(x|z) \right\} + D(Q(z) \parallel P(z)) + \log P(x)$$

$$\Rightarrow \log P(x) - D(Q(z) \parallel P(z|x))$$

$$= \mathbb{E}_{z \sim Q} \left\{ \log P(x|z) \right\} + D(Q(z) \parallel P(z))$$

we started from arbitrary dist for Q so
we pick $Q(z|x)$

$$\log P(X) - D[Q(z|X) \parallel P(z|X)] = \mathbb{E}_{z \sim Q} [\log P(X|z)] - D[Q(z|X) \parallel P(z)]. \quad (5)$$

the left hand side is the quantity that we want to maximize.

Question: Why we rewrite it like this?

$$\log P(X) - \mathcal{D}[Q(z|X) \| P(z|X)] = E_{z \sim Q} [\log P(X|z)] - \mathcal{D}[Q(z|X) \| P(z)]. \quad (5)$$

I mean which term we maximize and how?

if you want to know what to maximize
see what is your objective. $\log P(x)$ in this case

→ then see how you can maximize your objective.

why we can't get $P(z|X)$?

$$P(z|X) = \frac{P(X|z) P(z)}{P(X)}$$

impossible to compute

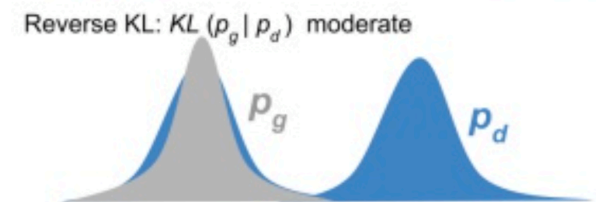
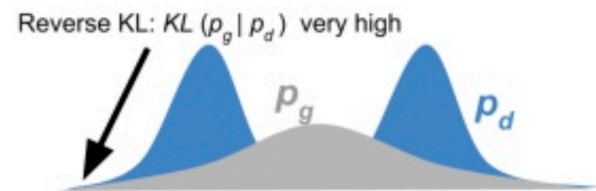
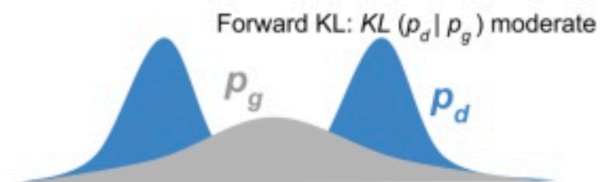
Forward KL : $KL(p||q) = \int p \log \frac{p}{q} dz$,

In this case, the model will try to avoid placing zero probability mass anywhere where $p > 0$ since this would lead to an exploding KL. This means it is safer to place non-zero q anywhere where plausibly $p > 0$ which leads to overestimation. This case is called zero-avoiding.

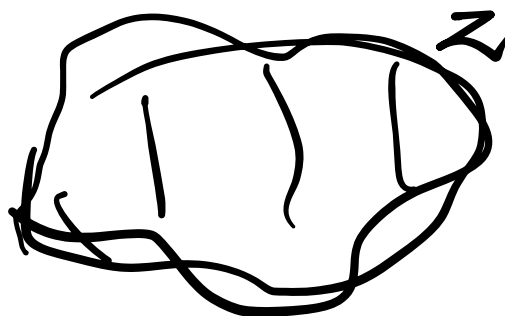
Reverse KL : $KL(q||p) = \int q \log \frac{q}{p} dz$,

In this case, the model will try to avoid situations where $q \approx 0$ and $p > 0$, thus it is safer to choose a mode and underestimate variance, rather than overestimate and risking to overshoot. This is called zero-forcing.

Assume p is true distribution and we want to approximate it with q .



$$P(x) = \int P(x|z, \theta) P(z) dz$$



$$\int P(x|z, \theta) P(z) dz$$

$$E_{Q_2} (P(x|z))$$

$$P(\theta | \mathcal{D})$$

$$Q(\gamma)$$

$$E_{z \sim Q} \{ P(\underline{x} | z) \} \approx \underline{P(x)}$$

$$\underline{P(z | x)}$$

$$Q(z) \leftarrow \underline{P(z | x)}$$

$$D \sim \{ Q(\underline{z}) \parallel \underline{P(z | x)} \}$$

$$E_Q \left\{ \log \frac{Q}{P} \right\}$$

$$\equiv \{ \log Q - \log P(Z|X) \}$$

$$\overline{E \{ \log Q \} - E(\log P(X|Z)) + \log P(Z)} \\ = 0$$

$$= D(Q(Z) \parallel P(Z)) + \log P(X|Z)$$